AIMirror: A Cognitive Architecture for Computational Identity Construction from Behavioral Evidence

Karthik T L
Department of Computer Science
AMC Engineering College
Bengaluru, India
karthiktl143@gmail.com

Manish N
Department of Computer Science
AMC Engineering College
Bengaluru, India
man18ish14@gmail.com

Nagaraj C G
Department of Computer Science
AMC Engineering College
Bengaluru, India
nagarajcg941@gmail.com

Niranjan C N
Department of Computer Science
AMC Engineering College
Bengaluru, India
cnniranjan72@gmail.com

Ms. Mala M
Assistant Professor, Department of Computer Science
AMC Engineering College
Bengaluru, India
mala.naik@amceducation.in

***Abstract*—Intelligent systems lack a coherent mechanism for constructing and reasoning about computational identity from behavioral data. Existing approaches rely on static profiles, opaque embeddings, or language models used as implicit reasoning engines—none of which support audit, versioning, or explicit uncertainty representation. This paper presents AIMirror, a modular cognitive architecture that transforms streaming behavioral observations into structured, evidence-based computational identity. The architecture comprises eleven specialized engines in two pipelines, governed by five invariant principles: the language model never reasons or decides; identity is read from immutable snapshots; every inference references supporting and countervailing evidence; all decisions use deterministic scoring; and components communicate via typed contracts. Core novelties include Behavior Objects as canonical behavioral units, an evidence engine computing net confidence from counter-evidence, identity snapshots with rollback, and cognitive planning without language model invocation. Because the resulting model is derived independently of any platform, we also use it to audit: comparing a platform's declared interests against the user's own behavioral evidence, separating interests the user sought from those delivered to them, and scoring the system's own claims against the user's verdicts, reported as calibration rather than accuracy. Corrections entered there are withdrawn from what the system subsequently asserts. We evaluate across 596 intent classification queries (macro F1 = 0.91, falling to 56% accuracy on phrasing held out from development), determinism verification, latency profiling, and measured ablations in which weakening a single safeguard is shown to discard every behavioral object produced from 800 events.**

***Index Terms*—cognitive architecture, computational identity, evidence-based reasoning, behavioral inference, self-model, identity snapshot, explainable AI, algorithmic accountability, confidence calibration**

I. Introduction

The ability to construct a computational representation of user identity from observed behavior is fundamental for intelligent interactive systems. Personalization, recommendation, digital wellbeing, and human-AI collaboration depend on modeling who the user is and how their preferences evolve [1]. Despite decades of research in user modeling, systematically transforming raw behavioral observations into an inspectable representation of computational identity remains largely open.

Existing approaches suffer from three interrelated limitations. First, user modeling relies on static features or matrix factorizations updated in batch, lacking identity tracking at the level of individual behavioral episodes [2]. Second, reasoning from behavioral signals to identity conclusions remains opaque inside neural network weights [3]. Third, platform-centric optimization maximizes engagement without providing insight into how behavioral data is interpreted [4].

Recent progress in retrieval-augmented generation [5] and large language models has renewed interest in memory-augmented systems. However, these approaches typically treat the language model as the central reasoning engine, delegating planning and decision-making to a stochastic process that is difficult to verify or reproduce [6], producing fluent but epistemically ungrounded outputs.

This paper takes a fundamentally different position: computational identity should be treated as a *first-class architectural construct* rather than an emergent property of language models or neural embeddings. AIMirror operationalizes this through a layered architecture in which evidence accumulation, identity evolution, cognitive planning, and language generation are explicitly separated. The language model appears exactly once, at the terminal stage, as a verbalizer of structured plans—never reasoning, deciding, planning, retrieving, or inferring.

The key contributions of this work are:

1) A modular cognitive architecture comprising eleven specialized engines operating on strongly-typed contracts,

with the language model restricted to terminal verbalization.
2) Behavior Objects as the minimum canonical behavioral unit, aggregating raw events into structured representations with lifecycle state (emerging, growing, stable, dormant, declining, dead).
3) An evidence propagation pipeline that collects supporting and countervailing evidence, computes net confidence, and surfaces conflicting observations for audit.
4) Identity Snapshots as immutable, versioned representations of computational identity, enabling consistent reads across conversational turns and rollback capability.
5) A Self-Model with explicit uncertainty representation, distinguishing strong beliefs from uncertain ones through counter-evidence tracking.
6) A rule-based cognitive planning layer performing intent classification, retrieval planning, reasoning mode selection, and response structuring entirely before language model invocation.
7) A deterministic decision layer that scores candidate content on six weighted dimensions, enforces topical diversity as a separate constraint, and records the score of every candidate for inspection.
8) An accountability layer that turns the identity model outward and inward: auditing a platform's stated claims about the user against behavioral evidence, distinguishing chosen interests from delivered ones, and calibrating the system's own confidence against user judgement, with corrections that bind on subsequent output.

The remainder of this paper is organized as follows. Section II reviews related work and presents a structured comparison. Section III presents the architectural design and rationale. Section IV presents the overall system architecture. Section V details each pipeline stage. Section VI describes the accountability layer. Section VII provides formal mathematical formulations. Section VIII presents evaluation results. Section IX discusses limitations, threats to validity, and future work. Section X concludes.

II. Related Work

Cognitive architectures such as SOAR [7] and ACT-R [8] model human cognition through production-rule memory and layered deliberative processing. AIMirror borrows their separation of memory tiers, but its object of study is different: it constructs a representation of one person from observed behaviour rather than a general model of cognition. User modeling has moved from stereotypes [1] through collaborative filtering [4] and matrix factorization [2] to neural methods [9]. Digital twin research applies similar ideas across manufacturing, healthcare and HCI [12], though for simulation rather than identity; work on computational identity [13] has explored the philosophical dimension without an implementable architecture.

Two further threads bear directly on the design. The explainable AI literature separates intrinsic interpretability from post-hoc explanation [3], and argues that consequential decisions warrant models that are interpretable by construction [10]; that argument motivates our decision to keep every inference inspectable rather than explaining it afterwards. Memory-augmented systems—retrieval-augmented generation [5] and recent persistent-memory agents [16]—inherit the episodic, semantic and procedural distinction from cognitive science [11]. AIMirror keeps observations, consolidated objects, evidence, inferences, reflections, goals and an importance-ranked recall index in separate typed stores, and—unlike these systems—selects among them with a cost-bounded planner that never calls a language model.

The closest contrast is with agent frameworks that place a language model at the centre of control. ReAct [17] interleaves reasoning and action as generations; AutoGPT [18] and comparable frameworks extend this to autonomous multi-step execution. The flexibility is real, and so is the cost: nondeterministic behaviour, opaque reasoning paths, and fabricated intermediate steps [6]. AIMirror takes the opposite position, and the rest of this paper is largely an account of what that costs and what it buys.

A. Comparative Analysis

Table I compares AIMirror against representative systems across six architectural dimensions. No existing system simultaneously provides identity evolution, intrinsic explainability, multi-memory architecture, evidence tracking with counter-evidence, LLM-free reasoning, and immutable identity snapshots.

TABLE I
Architectural comparison across representative systems and dimensions. Checkmark (✓) indicates full support; (○) indicates partial support.

System	Identity	Explain	Memory	Evidence	No LLM	Snapshot
SOAR [7]	○	✓	✓	–	✓	–
ACT-R [8]	–	✓	✓	–	✓	–
RAG [5]	–	○	–	–	–	–
Neural CF [9]	–	–	–	–	–	–
Digital Twin [12]	–	○	○	–	–	–
XAI Methods [10]	–	✓	–	–	–	–
AIMirror	✓	✓	✓	✓	✓	✓

B. Research Gap

Existing systems fail to provide a complete, verifiable pipeline from raw behavioral observations to structured computational identity with: (a) explicit evidence provenance and counter-evidence tracking, (b) immutable identity snapshots enabling consistent multi-turn interaction, (c) a self-model with quantified uncertainty, (d) cognitive planning independent of language model reasoning, and (e) deterministic decision optimization preceding language generation. AIMirror addresses this gap through a deliberately layered architecture where each stage produces verifiable, inspectable outputs.

III. Architectural Design

The obvious objection to this work is that it could be assembled from existing parts. Three decisions resist that reading. First, identity is *constructed* as an explicit output—observations, evidence, inferences, sub-profiles, snapshot—rather than inferred as a byproduct of some downstream objective and left implicit in parameters. The architecture further separates *Identity*, what the behavioural data shows, from the *Self Model*, what the system believes, and models the gap between them through counter-evidence and uncertainty rather than collapsing it.

Second, the language model is confined to terminal verbalization, which two ordering rules enforce: planning precedes retrieval, so sources and reasoning mode are fixed before any data is fetched; and decision precedes verbalization, so the model formats content that has already been selected rather than choosing what to say. Third, query-time reads come from frozen snapshots, so an identity under construction cannot mutate mid-conversation.

The component choices follow from these commitments. Behavior Objects give heterogeneous events a uniform interface with lifecycle state. The intent classifier is rule-based, trading some accuracy for plans that are identical on identical input. The Decision Engine uses fixed weights rather than learned ranking for the same reason: a score that can be recomputed by hand is one a user can contest.

IV. System Architecture

AIMirror comprises eleven specialized engines arranged in two pipelines: the *ingestion pipeline* (event to identity) and the *query pipeline* (query to response). Figures 1 and 2 provide focused views of each pipeline.

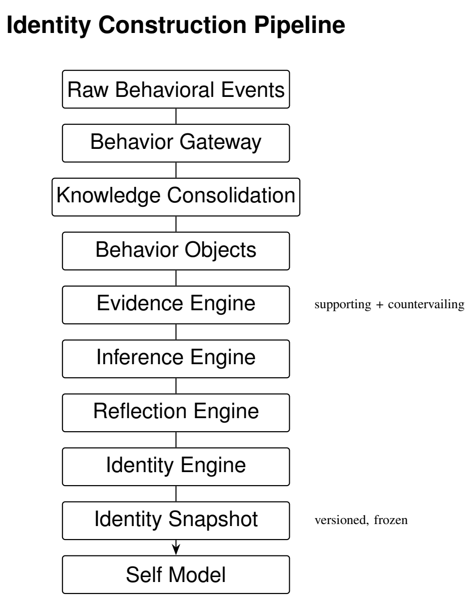

Fig. 1. Identity construction pipeline: raw behavioral events are normalized, consolidated into behavior objects, processed through evidence and inference engines with counter-evidence tracking, assembled into a structured identity, frozen into immutable snapshots at evolutionary thresholds, and exposed through a self-model with uncertainty quantification.

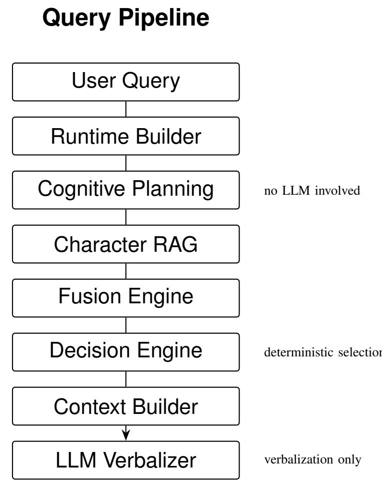

Fig. 2. Query pipeline: user queries are processed through runtime construction, language-model–free cognitive planning, retrieval-augmented generation, multi-source fusion, deterministic decision scoring, context assembly, and terminal language model verbalization. The language model is invoked exactly once, at the final stage.

A. Architecture Principles

The architecture is governed by five invariants: (1) the LLM operates exclusively at the final stage, never performing reasoning, retrieval, or decisions; (2) all user-facing operations read from frozen Identity Snapshots, never the live identity; (3) every inference and belief must reference supporting and countervailing evidence; (4) the Decision Engine uses deterministic weighted scoring; and (5) all components communicate through strongly-typed models.

V. Pipeline Stages

A. Ingestion: from events to identity

Figure 1 shows the path from raw behavioural events to the self-model. The Behavior Gateway normalises heterogeneous platform events into typed contracts, after which Knowledge Consolidation groups them into clusters by topical and temporal proximity. Each surviving cluster becomes a *Behavior Object*: the canonical unit of the architecture, carrying its topic, contributing creators, aggregate engagement statistics, and a lifecycle state drawn from emerging, growing, stable, dormant, declining and dead. Working at this granularity rather than on individual events is what makes later stages inspectable, because every downstream claim can name the objects it rests on.

The Evidence Engine then collects evidence across five dimensions—topical, temporal, creator, interaction and cross-platform—recording countervailing observations alongside supporting ones and computing a net confidence that a single accumulation counter cannot express. The Inference Engine applies rule-based condition–action pairs in priority order to derive higher-level conclusions, each recording its type, quality metrics, supporting evidence and suggested actions. The Reflection Engine summarises each consolidation over the span it covers, earliest to latest observation, labelled daily, weekly or monthly by that measured length rather than by an assumed schedule.

The Identity Engine assembles nine sub-profiles—behaviour, interest graph, creator graph, learning style, attention, exploration, consistency, habit and motivation—each with a confidence computed from the evidence that profile alone rests on—the interest graph from topic behaviours, the creator graph from those naming a creator, attention from those carrying watch statistics, motivation from inferences—as $\min(0.95,\ v \cdot r)$, for volume v saturating at twenty observations and recency r halving every thirty days from the newest support. Multiplied rather than averaged, since abundant year-old observations describe not a half-confident profile but a person who may no longer exist; the ceiling sits below one because these say how much was seen and how lately, not whether it was right. The identity-wide figure renormalises over the components that exist rather than substituting a neutral value for absent ones, and is capped by v. The Snapshot Manager watches the active identity for shifts exceeding an L_2 distance of 0.30 and, when one occurs, writes an immutable versioned snapshot. Since invariant 2 already routes user-facing reads through a frozen snapshot, rollback is the choice of *which* one is frozen: a person who does not recognise where the model has drifted sends reads to an earlier snapshot, and nothing is deleted or rewritten. Overwriting the live identity instead would not survive, since construction is from scratch on each ingest, so the control would appear to work while quietly undoing itself. All seventeen dimensions are scaled to $[0, 1]$ before the distance is taken; sixteen are already bounded scores, while mean attention span is a duration and is divided by a 300-second cap. Mixing an unbounded term into the norm makes the threshold meaningless, as Section IX reports. Distinct from all of this is the Self Model, which holds what the system *believes* rather than what the data shows: each belief carries a type, a confidence, an uncertainty counter and references to both supporting and countervailing evidence, and is classified STRONG only when it is both well supported and not contested by its own observations, UNCERTAIN otherwise (Eq. (5)). That distinction is what lets the verbalizer hedge honestly instead of asserting everything with equal force.

B. Query: from question to answer

Figure 2 shows the query path. The Runtime Builder assembles a working context from the active snapshot; the Cognitive Planning Layer then classifies intent, plans retrieval, selects a reasoning mode and fixes the response structure—all through deterministic rules, with no language model involved. Character RAG retrieves across the five memory types (episodic, semantic, behavioural, goal and reflection), each with its own storage and retrieval strategy, with the planner deciding which to query. The Decision Engine scores candidate content on six weighted dimensions, caps how many facts any one topic may contribute, and selects what may be said, so low-confidence or contradictory material never reaches the generator. Only then does the Context Builder assemble a structured plan for the LLM Verbalizer, whose sole task is to render that plan as fluent prose.

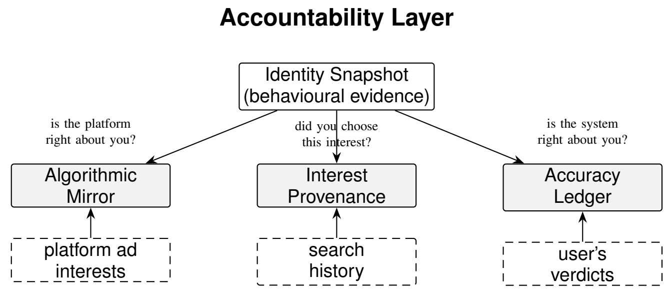

Fig. 3. The accountability layer. One identity model, derived independently of any platform, is compared against three different references: the interests a platform declares about the user, the searches that show what the user went looking for, and the user's own verdicts on the system's claims. Corrections entered at the ledger are withdrawn from what the system subsequently asserts.

VI. Accountability Layer

An identity model that only describes a user is of limited interest. Because AIMirror derives its model independently of any platform, it can also be used to check claims that are ordinarily unverifiable, and Figure 3 shows the three surfaces this produces, each set against an external reference. A fourth needs no external reference at all.

Auditing the platform. Advertising interests exported from a user's Meta or Google archive are compared against the twin's own evidence and sorted into corroborated, unsupported, and not comparable. The third category matters: a claim such as "frequent travellers" is demographic and a watch history can neither confirm nor refute it, so filing it as a failure would overstate the platform's error. Below a coverage floor the report withholds a verdict entirely rather than judging a platform on a fragment of a user's history.

Auditing provenance. For each topic the system separates deliberate acts—searches, which require the user to have gone looking—from exposure. Agency is the ratio of the two, capped at one, and a topic is reported as fed, mixed or chosen. Reporting a chosen interest as fed is the more harmful error, since it tells someone an interest was installed in them when they sought it out, so the matching that connects a search to a topic is deliberately permissive; the corroboration test above, where a false match instead launders an unevidenced claim, uses a stricter threshold.

Auditing itself. Users mark each claim the system makes as right, wrong or unsure. The ledger reports *calibration* rather than accuracy: accuracy alone is trivially maximised by never claiming anything confidently, whereas the useful question is whether claims made at confidence c are correct about c of the time. Rates are withheld for any confidence band with fewer than ten answers and no overall verdict is given below twenty, with Wilson intervals throughout, since the normal approximation reports certainty at exactly the sample sizes this feature lives in. A Brier score accompanies the rates, because it penalises confident errors in a way a percentage cannot.

Auditing itself without asking. The ledger above needs the user to answer; this needs nothing from anyone. Every piece of evidence records the observations contradicting it alongside those supporting it (Eq. (2)), so each claim carries its own dissent and can be ranked by how contested it is rather than how confident—the ordering every other surface uses. The report names the content behind each disagreement, since stating that nine of twenty-four observations argue against a claim without showing the nine asks to be trusted about being untrustworthy. Because the partition is computed from watch time alone, it applies to every claim automatically, including the ones nobody has been asked about, and it can report that a claim has no dissent at all.

Corrections bind. A claim the user marks wrong is withdrawn from everything the system asserts, while remaining visible and reversible where the reasoning is shown—hiding it would make the correction untraceable. Since the pipeline regenerates inferences on every ingest, a claim is identified by a fingerprint of its content rather than by row identity, so a correction survives regeneration. Finally, each claim states the threshold it turns on, distinguishing conditions a person could act on from those that merely reflect how much data exists.

VII. Formalization

A. Behavior Object

A Behavior Object b is defined as a tuple (Eq. (1)):

$$b = \langle \mathcal{T}, \mathcal{C}, \mathcal{S}, \ell, \alpha, \gamma, \delta \rangle \tag{1}$$

where $\mathcal{T}$ is the topic hierarchy, $\mathcal{C}$ is the set of associated creators, $\mathcal{S}$ are the four statistical sub-models (engagement, watch, temporal, trend), $\ell \in \{\text{EMERGING}, \text{GROWING}, \text{STABLE}, \text{DECLINING}, \text{DORMANT}, \text{ARCHIVED}\}$ is the lifecycle state, and $\alpha, \gamma, \delta \in [0, 1]$ are the importance, confidence, and stability scores.

The state is evaluated when asked rather than fixed at consolidation: a behaviour someone has abandoned never appears in another batch, so a state written once cannot be revised for precisely those needing retirement. Silence is judged against the behaviour's own rhythm, $\Delta_{\text{last}} \geq \max(14 \text{ days}, \ 3/\phi_b)$ for historical daily frequency ϕ_b, since a topic recurring daily and one recurring monthly are not equally absent after a fortnight; the floor stops a burst reading as abandonment two days later. Direction comes from the share of occurrences in the recent half of the lifespan, not from `growth_rate`, which is $|b|/\Delta_{\text{age}}$ and positive by construction.

B. Evidence Confidence

For an evidence item e with supporting observations S_e and countervailing observations C_e, the net confidence $\hat{\gamma}_e$ is (Eq. (2)):

$$\hat{\gamma}_e = \gamma_e \cdot \max\left(0, \ \frac{|S_e| - |C_e|}{|S_e| + |C_e|}\right) \tag{2}$$

where γ_e is the evidence confidence, a function of how much was observed. The factors answer different questions and are kept apart: γ_e states how much was seen, the balance term whether it pointed one way, so a claim resting on ample but self-contradictory data cannot hide behind its sample size. Evidence split evenly is fully offset rather than halved, and $\hat{\gamma}_e$ is withheld below five observations, where one skip moves the balance from $+1$ to -1.

An observation counts against a claim when the content was watched for under 0.40 of that user's median watch time—relative, since an absolute cut in seconds would classify two users with different habits identically. On the deployed corpus this marks 19–20% of events as skipped, against 13–15% at 0.25 and 22–23% at 0.50, so the reading does not turn on the exact value. A user who skips nothing yields no counter-evidence, the property that makes its absence meaningful; taking each history's bottom quartile instead would manufacture counter-evidence for everyone.

C. Identity Evolution and Snapshots

Given identity state I_t at time t, the evolution function Φ maps the current identity, new evidence E_t, accumulated Behavior Objects B_t, and recent reflections R_t to the next identity state (Eq. (3)):

$$I_{t+1} = \Phi(I_t, E_t, B_t, R_t) \tag{3}$$

A snapshot is created when the identity shift exceeds the evolution threshold τ_{identity} (Eq. (4)):

$$\|I_{t+1} - I_t\|_2 > \tau_{\text{identity}} \tag{4}$$

where $\|\cdot\|_2$ is computed over the confidence-weighted feature vector of each sub-profile.

D. Self-Model Belief Status

A belief b in the Self Model is characterized by (Eq. (5)):

$$\text{status}(b) = \begin{cases} \text{STRONG}, & \sigma_b \geq 0.7 \ \wedge \ \nu_b > \frac{1}{3} \\ \text{UNCERTAIN}, & \text{otherwise} \end{cases} \tag{5}$$

where σ_b is belief strength and $\nu_b = (|S_b| - |C_b|)/(|S_b| + |C_b|)$ is the balance of observations beneath it. Requiring *no* counter-evidence would be more conservative and we do not adopt it: nearly every real claim accumulates some skipped observations, so that bar makes STRONG unreachable and collapses the distinction it draws. One observation in three dissenting is the line, leaving both statuses attainable—on an account with a 19% skip rate, one belief of four was STRONG, ν_b spanning 0.11 to 0.63. The predicates are complementary, so no belief is both settled and open.

E. Decision Score and Selection

The decision score $D(f)$ for a candidate fact f is the weighted sum in Eq. (6), clamped to $[0, 1]$:

$$D(f) = \min\left(1, \ \sum_{d \in \mathcal{D}} w_d \cdot \sigma_d(f)\right) \tag{6}$$

where $\mathcal{D}$ is the set of six scored dimensions with weights w_d: confidence (0.25), identity alignment (0.20), recency (0.15), goal alignment (0.15), evidence strength (0.10) and stability (0.10). Each $\sigma_d(f) \in [0, 1]$ is that dimension's score for the fact. Topical diversity is enforced separately, as a hard cap of $k_{\text{topic}} = 5$ facts per topic applied after scoring, rather than as a term in the sum—a weighted boost cannot guarantee that no single topic dominates, whereas a cap can. The selection set $\mathcal{F}^*$ follows in Eq. (7):

$$\mathcal{F}^* = \{f \in \mathcal{F} : D(f) \geq \tau_{\text{conf}} \wedge \text{rank}(f) \leq k\} \tag{7}$$

where $\mathcal{F}$ is the set of candidate facts, $\tau_{\text{conf}} = 0.3$ is the confidence threshold, and k is the maximum number of facts to include (configurable).

F. Intent Classification

Given a query q, the intent planner computes a score s_i for each intent type i (Eq. (8)):

$$s_i = \frac{|\{p \in P_i : p(q) = \text{true}\}|}{|P_i|} \tag{8}$$

where P_i is the set of regex patterns for intent i. Ties are resolved by a priority function $\pi(i)$ (Eq. (9)):

$$i^* = \arg\max_i \, (s_i, \pi(i)) \tag{9}$$

with lexicographic ordering on the pair $(s_i, \pi(i))$.

G. Confidence Accumulation

The overall confidence Γ of the system's output for a given query is the product of confidences across n pipeline stages:

$$\Gamma = \prod_{j=1}^{n} \gamma_j \tag{10}$$

where γ_j is the confidence of stage j. This multiplicative formulation ensures low confidence at any stage propagates to the final output, providing a natural mechanism for expressing uncertainty.

H. Implementation

AIMirror is implemented as an asynchronous Python backend using FastAPI with PostgreSQL and pgvector for hybrid semantic-temporal vector search. Embeddings use Sentence-BERT's `all-MiniLM-L6-v2` [15] producing 384-dimensional vectors. The database schema comprises relational tables for all cognitive state—behavior objects, evidence carrying its countervailing observations as indexed JSONB arrays, inferences with rule provenance, nine-sub-profile identities, immutable snapshots, self-models with belief arrays, and stores for memories, reflections, and goals—enforcing referential integrity across stages for full provenance tracing. Hybrid retrieval combines cosine similarity (weight 0.7) with PostgreSQL full-text ranking (weight 0.3), using IVFFLAT indexing for approximate nearest neighbour search; a separate recency-decayed search serves time-sensitive lookups. Cosine rather than Euclidean distance matters despite both ordering unit-length embeddings identically, since only the former shares a scale with the keyword term it is blended against. The query pipeline stages (RuntimeBuilder → PlannerOrchestrator → Retriever → MemoryRanker → FusionEngine → DecisionEngine → ContextBuilder → LLMVerbalizer) each log inputs, outputs, confidence, and timing for auditability. The API exposes `POST /ingest` (asynchronous) and `POST /query` (synchronous) endpoints, with fail-open for ingestion and fail-closed for queries.

VIII. Evaluation

A. Pipeline Determinism

We verified determinism by executing the full query pipeline 100 times on an identical snapshot with an identical query. Both planning and Decision Engine outputs produced identical cryptographic hashes across all runs (100/100 identical outputs), confirming the elimination of nondeterminism from all stages preceding verbalization.

B. Intent Classification Accuracy

We constructed a labeled dataset of 596 natural language queries balanced across 11 intent types (53–55 per class), combining 110 manually curated queries with 486 synthetic variations from 10 syntactic patterns per class. Aggregated, the rule-based classifier reaches 90.44% accuracy and macro F1 0.9063.

That aggregate is not a generalisation estimate, though an earlier version of this work reported it as one. Its halves differ completely—100% on all 110 curated queries against 88.3% on the synthetic half—because the patterns were written from the curated queries, and a rule set scores perfectly against its own examples. Inspecting the synthetic failures spent that half too. We therefore wrote 36 further queries by hand, in phrasings neither set contains, and read no individual failure of theirs until the work was finished: accuracy there is 55.6%. A regex classifier recognises the constructions it was given and degrades on unrehearsed phrasing, so the aggregate measures the benchmark's composition as much as the classifier.

Two changes account for the improvement from an initial 81.21%, both diagnosed from the synthetic half. Identity questions were matched on sentence frame rather than subject—"describe my values" is an identity question in an informational frame—which raised identity recall from 0.35 to 0.67 synthetic and 0.38 to 1.00 held out. And 55 of 98 synthetic errors matched no pattern in any class, landing in `unknown` by default rather than by argument, so ordinary constructions absent from the rules were added. Held-out accuracy moved 30.6% → 44.4% → 55.6% across the two, and nine of the sixteen residual failures still match nothing, so coverage rather than discrimination remains the limit. Because the reading also selects the retrieval directives, a misreading answers from the wrong stores; each answer therefore reports its reading and offers the alternatives, and choosing one re-plans the query under it while keeping the topics already extracted. It reports separately that no pattern matched, which is the state worth flagging: 9.1% correct held out against 76.0% for the rest.

C. Baseline Comparison

Against a language model answering directly from raw identity data and against standard retrieval-augmented generation, the differences are those Table I records. They are structural rather than incremental: fabrication is excluded before generation rather than mitigated after it, consistency holds across sessions because reads come from a frozen snapshot, and pre-verbalization output is byte-identical across runs.

D. Component Latency Profile

Averaged over 10 query pipeline executions, every cognitive stage completes in under 3 ms—memory ranking (2.3 ± 0.4) and runtime load (2.0 ± 0.3) the slowest, the remaining five at or below 1.1 ms—for a cognitive total of 7.8 ± 1.4 ms. End-to-end is approximately 3.5 s, dominated by database connection pool initialization on cold start rather than by any reasoning stage.

E. Confidence Product Validation

Across all 596 queries the mean Γ (Eq. (10)) was 0.342 (SD = 0.187), higher for correctly classified queries (0.418) than misclassified ones (0.124)—a difference in means reported without a hypothesis test, so indicative rather than established. The right-skewed distribution (23% above 0.5, 12% below 0.1) shows low-confidence stages propagating to low-confidence outputs.

F. Architecture Invariant Validation

Six invariants are checked programmatically rather than by convention: the Runtime Builder is the sole entry point for query-time loading, every request produces a structured plan, retrieval never bypasses planning, the Decision Engine sits strictly between Fusion and Context construction without modifying Fusion's output, and the language model performs no reasoning. All held across all test executions.

Fused retrieval combines up to nine sources with a mean set size of 42.3 items (SD = 11.7) and 31% mean deduplication; the Decision Engine then reduces 14–50 candidate facts to typically 1–4 before verbalization, eliminating chiefly on low confidence (71% of removals) and diversity constraints (19%).

IX. Discussion

A. What the separation buys

Each stage can be checked on its own. Confining the language model to terminal verbalization removes the main source of nondeterminism; counter-evidence tracking makes contradiction visible; snapshots give multi-turn interaction a fixed reference. The contrast with agent frameworks [17], [18] is one of purpose rather than quality: those systems pursue open-ended goals and keep their reasoning inside the model, while AIMirror answers within a fixed pipeline and can name the observations behind every claim. The two are complementary.

B. Measured effects of weakening a safeguard

Rather than reason about what removing a component would do, we weakened five safeguards on the deployed instance and measured what the pipeline produced. Table II reports the results. Two are worth drawing out.

Removing a single bounds check is enough to destroy the entire model. Recency is scored as $1 - \Delta_{\text{days}}/30$, and Python's `timedelta.days` floors toward negative infinity, so an event timestamped even seconds in the future yields -1 days and a score of 1.033—outside the $[0, 1]$ bound the Behavior Object declares. Construction raises, and because consolidation caught exceptions for the whole step rather than per cluster, every behaviour object for that run was discarded: 800 events produced none, and therefore no inferences and no identity. The failure was logged but indistinguishable downstream from an account with no data. Clock skew on a client produces the same input.

Second, evidence that is not scoped to a claim is worse than no evidence. Before each rule declared which entities it had reasoned over, the inference engine attached the eight most active behaviour objects to every inference regardless of rule; across production, 10 of 10 users had exactly one distinct creator set spanning all of their claims. Displayed as justification, that reads as "this is why" while bearing no relation to the conclusion.

TABLE II
MEASURED EFFECT OF WEAKENING EACH SAFEGUARD, ON A DEPLOYED INSTANCE. CORPUS: 195 BEHAVIOUR OBJECTS AND 6,451 CAPTIONED EVENTS ACROSS 10 ACCOUNTS, OF WHICH MOST ARE SYNTHETICALLY SEEDED.

Safeguard weakened	**Measured effect**	**Magnitude**
Recency bounds check	Behaviour objects discarded	800 events → 0 objects
Per-cluster error isolation	One bad cluster voids the run	all topics lost, not one
Rule-declared evidence scope	Same evidence under every claim	10/10 users, 1 distinct set
Corroboration threshold (0.45 → 0.30)	Unsupported claims corroborated	4/12 false vs 1/12
Topic-scoped keyword match	Over-broad classification	45% vs 9% of objects
Identity-vector scaling	Norm dominated by one term	shift 118.2 vs bound 4.12

The scaling row is the sharpest case of a threshold that looks principled and is not. Sixteen of the seventeen identity dimensions are scores bounded in $[0, 1]$; mean attention span is a duration in seconds and was entering the norm unscaled. Eq. (3) was therefore not measuring identity shift but a change in seconds, since a term ranging over hundreds cannot be outweighed by sixteen that differ by at most one. Driving the pipeline through a deliberate persona change—deep technical viewing at 120–240 s replaced by shallow entertainment at 3–12 s—recorded a shift of 118.2, where the geometric bound on seventeen scaled dimensions is $\sqrt{17} = 4.12$. Against a threshold of 0.30 a snapshot was warranted whenever mean attention moved by a third of a second, which is how one account accumulated fifteen consecutive snapshots with identical contents.

The corroboration row concerns the threshold at which a platform's stated interest counts as supported by behaviour. Against 24 hand-labelled claim–topic pairs, genuine matches score no lower than 0.515 and spurious ones no higher than 0.514; at 0.30 the system corroborated "Automotive" against `tech` (0.424). The pairs were labelled by the authors, so 0.45 is a considered default rather than a tuned optimum.

C. Limitations

The alignment module implements rule-based scoring only, without reinforcement learning. The architecture has not been evaluated in a controlled user study, and the corpus behind Table II is dominated by seeded accounts, so the magnitudes are indicative rather than population estimates. Ingestion is currently single-source, and commits events and the state derived from them in one request with no resumption if that request is interrupted: 13 of 35 accounts on the deployed instance, and 7 of 15 seeded ones, held events that nothing had consolidated, and were therefore unanswerable while appearing populated. Such an account now reports that state rather than an empty retrieval result, which had made it indistinguishable from a failure. A related bound applies to clustering, which requires three co-occurring events: applied to one request in isolation it discarded 19–50% of an account0̆027s topics, almost all of them creators, and now runs over a window of recent stored events instead. Finally, the learning-orientation rule classifies subject matter rather than instructional intent: `#coding` says what content is about, not whether it teaches. Captions carry the better signal (24% of 6,451 events contain instructional phrasing), but calibrating on it needs a corpus that is not mostly synthetic.

D. Threats to Validity

The held-out set that produces the 55.6% figure is 36 queries written by an author who had seen the classifier's failures on the other sets, so it is not an unbiased sample either: it establishes the direction rather than the magnitude. Having now been examined, it cannot serve as held-out evidence for a further change. Determinism was verified on one snapshot with fixed queries. Evaluation is single-deployment. Confidence scores are internally consistent but not calibrated against any external ground truth, so their correspondence to psychologically meaningful constructs is unestablished. The source, migrations, the evaluation sets, the evaluation scripts and deployment documentation are released together; every stage before verbalization is deterministic, so runs reproduce without seed control.

E. Future Work

The most useful next steps are empirical rather than architectural: a controlled user study measuring whether the accountability surfaces change what people believe about their own behaviour, and a corpus of real rather than seeded accounts, without which several thresholds in this paper remain considered defaults. Beyond that: multi-source ingestion, reinforcement learning in place of rule-based alignment scoring [14], and benchmarking the two-pipeline design at scale.

X. Conclusion

This paper presented AIMirror, a modular cognitive architecture that treats computational identity as a first-class construct—built through evidence accumulation, identity evolution, cognitive planning and controlled language generation—rather than as a byproduct of LLM reasoning or static feature engineering. Separation of concerns runs across eleven engines with the language model confined to terminal verbalization. The distinctive elements are Behavior Objects with lifecycle tracking, an evidence engine that records what contradicts a claim alongside what supports it, immutable snapshots, a self-model with explicit uncertainty, and planning without model involvement.

Evaluation shows deterministic pipeline behaviour (100/100 identical outputs), macro F1 of 0.91 across 596 intent queries and 56% accuracy on unrehearsed phrasing held out from development, sub-millisecond component latencies, invariant compliance, and measured ablations in which weakening a single bounds check discards every behavioural object produced from 800 events. Validation through a user study remains outstanding, and several thresholds await a corpus that is not mostly seeded. The reframing of identity from emergent artifact to constructed entity—grounded in evidence, versioned through snapshots, and answerable to the observations that contradict it—is the core architectural contribution.

References

[1] A. Kobsa, "Generic user modeling systems," *User Model. User-Adapt. Interact.*, vol. 11, no. 1–2, pp. 49–63, 2001.
[2] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, pp. 30–37, 2009.
[3] Z. C. Lipton, "The mythos of model interpretability," *Queue*, vol. 16, no. 3, pp. 31–57, 2018.
[4] P. Resnick and H. R. Varian, "Recommender systems," *Commun. ACM*, vol. 40, no. 3, pp. 56–58, 1997.
[5] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 9459–9474, 2020.
[6] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.
[7] J. E. Laird, A. Newell, and P. S. Rosenbloom, "SOAR: An architecture for general intelligence," *Artif. Intell.*, vol. 33, no. 1, pp. 1–64, 1987.
[8] J. R. Anderson *et al.*, "An integrated theory of the mind," *Psychol. Rev.*, vol. 111, no. 4, pp. 1036–1060, 2004.
[9] X. He *et al.*, "Neural collaborative filtering," in *Proc. WWW*, 2017, pp. 173–182.
[10] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, no. 5, pp. 206–215, 2019.
[11] E. Tulving, "Episodic and semantic memory," in *Organization of Memory*, E. Tulving and W. Donaldson, Eds. New York, NY, USA: Academic Press, 1972, pp. 381–403.
[12] F. Tao *et al.*, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, 2019.
[13] L. Floridi, "The informational nature of personal identity," *Minds Mach.*, vol. 21, no. 4, pp. 549–566, 2011.
[14] J. Schulman *et al.*, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.
[15] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.
[16] J. S. Park *et al.*, "Generative agents: Interactive simulacra of human behavior," in *Proc. UIST*, 2023, pp. 1–22.
[17] S. Yao *et al.*, "ReAct: Synergizing reasoning and acting in language models," in *Proc. ICLR*, 2023.
[18] AutoGPT, "AutoGPT: An autonomous GPT-4 experiment," GitHub repository, 2023. [Online]. Available: https://github.com/Significant-Gravitas/AutoGPT